SMART DISTAP · CHOY SUM LIGHT STRESS

Results

Classifying light stress in choy sum from three-band images

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THE DATA

48 three-band images of choy sum

4

Light conditions

White light (healthy control), high light, low light, shade

4

Time points

4 hours, then days 1, 2 and 3

3

Plants

Per condition and time point (P1, P2, P3)

The task: given one image, say which of the four conditions the plant was under. Guessing at random scores **0.25**.

THE METHOD

How the best model works, in five steps

1

Normalise

Divide each image by that day's healthy-plant average, so only stress differences remain

2

Cut

Cut the plant into about 9,000 overlapping patches

3

Describe

A pretrained ResNet-50 turns each patch into numbers

4

Label

A small classifier names a condition for each patch

5

Vote

The most common label among the patches is the plant's answer

Patch and stride	Patch: a 224 × 224 pixel square of the 608 × 968 image. Stride: each patch starts 4 pixels from the last
Fair test	Train on two plants, test on the third it has never seen; each plant takes a turn
Repeated runs	**15 runs** = 3 test plants × 5 trainings from different random starts. Each training lasts 60 epochs
Being tuned	These settings were fixed by hand. Bayesian optimisation is now tuning patch size, stride, epochs and learning rate

KEY RESULTS

0.96

Is the plant stressed?

Correct 96% of the time: 173 of 180 stressed plants caught, 58 of 60 healthy plants left alone

0.69

Which light condition?

Correct 69% of the time, against 25% by guessing. High light is always right; low light and shade get mixed up

Best model: ResNet-50 patches (stride 4) + softmax, plant vote, on plants not seen in training

EVERY TECHNIQUE COMPARED

The best techniques name the condition correctly 69% of the time

Technique	Accuracy	Precision	Recall	F1
ResNet-50 patches (stride 4) + softmax, stressed vs control (2 classes)	**0.96**	0.94	0.96	**0.95**
ResNet-50 patches (stride 56) + softmax, plant vote	0.69	0.65	0.69	0.65
ResNet-50 patches (stride 4) + softmax, plant vote	0.69	0.61	0.67	0.62
ResNet-50 layers 2+3+4 + SVM	0.69	0.68	0.69	0.68
SVM, whole plant	0.69	0.67	0.67	0.67
Tuned SVM (Bayesian optimisation)	0.65	0.64	0.65	0.64
PLS-DA, pixels, majority vote	0.62	0.44	0.58	0.50
ResNet-50 patches + softmax, single patches	0.61	0.56	0.57	0.57
PLS-DA, whole plant	0.50	0.41	0.54	0.46
ResNet-50 last layer + SVM	0.44	0.50	0.48	0.48

All time points (4 h to day 3). Top row: 2 classes (healthy vs stressed). Others: 4 conditions, chance 0.25

CONFUSION MATRIX · ALL TIME POINTS

Most mistakes are low light and shade mixed up

Actual ↓	White	High	Low	Shade	F1
White light	**58**	0	2	0	0.93
High light	0	**60**	0	0	0.82
Low light	6	15	**8**	31	0.19
Shade	1	11	14	**34**	0.52

Rows are the real condition, columns the model’s answer; bold is correct. 60 tests per condition

0.69

Naming the exact condition

0.96

Healthy vs stressed: a low light plant called shade still counts as stressed

Of 60 low light tests, 31 were called shade: both mean less light, so they look alike. The next slide splits this by time point.

ResNet-50 patches + softmax, plant vote. 60 = 3 plants × 4 time points × 5 runs

CONFUSION MATRIX · BY TIME POINT

The same pattern at every time point

4 hours · accuracy 0.70 · healthy vs stressed 0.97

Actual ↓	White	High	Low	Shade
White light	**13**	0	2	0
High light	0	**15**	0	0
Low light	0	0	**0**	15
Shade	0	0	1	**14**

Day 1 · accuracy 0.67 · healthy vs stressed 0.98

Actual ↓	White	High	Low	Shade
White light	**15**	0	0	0
High light	0	**15**	0	0
Low light	1	4	**0**	10
Shade	0	0	5	**10**

Day 2 · accuracy 0.70 · healthy vs stressed 1.00

Actual ↓	White	High	Low	Shade
White light	**15**	0	0	0
High light	0	**15**	0	0
Low light	0	1	**8**	6
Shade	0	11	0	**4**

Day 3 · accuracy 0.60 · healthy vs stressed 0.90

Actual ↓	White	High	Low	Shade
White light	**15**	0	0	0
High light	0	**15**	0	0
Low light	5	10	**0**	0
Shade	1	0	8	**6**

ResNet-50 patches + softmax, plant vote. 15 tests per condition = 3 plants × 5 runs. Bold = correct

TUNING · FIRST RUN

Smaller 64-pixel patches lift accuracy from 0.69 to 0.75

Measure	224 px, fixed	64 px, tuned
Which of 4 conditions	0.69	**0.75**
F1 (average)	0.65	**0.72**
Low light F1	0.19	**0.42**
Shade F1	0.52	**0.72**
At 4 hours	0.70	**0.83**
Healthy vs stressed	0.96	0.94

Patch size agrees

All 3 test plants chose 64 × 64 pixel patches, stride 4

Training settings do not

Learning rate, optimiser and epochs differ per plant; early stopping often picks 1 to 10 epochs

Preliminary

Laptop run, 30 trials, 300 patches per image; GPU run to confirm

ResNet-50 patches + softmax, plant vote. Nested Bayesian optimisation: the test plant never influences the chosen settings

FINDINGS

What improved accuracy, and what did not

Improved

- **Comparing with healthy plants each day:** 0.38 to 0.62 for pixel PLS-DA
- **One vote per plant:** 0.61 for single patches to 0.69 per plant
- **Middle ResNet layers:** 0.69, against 0.52 for the last layer alone
- **Smaller patches (64 pixels):** 0.69 to 0.75, first tuning run

No change

- **Patches closer together (stride 112, 56, 4):** all 0.69 to 0.75
- **Tuning SVM and PLS-DA settings:** no better than the defaults
- **Laptop vs Colab GPU:** same results, so they are reproducible

Takeaway: the limit is the data, not the settings. Three plants per condition, and low light and shade look alike.

NEXT STEPS

Tune the settings, then add kailan and Raman

1. **Confirm the tuning on the GPU.** First run chose 64-pixel patches (0.75). Rerun with more trials and patches, plus loss curves.

2. **Separate low light from shade.** Fisher analysis to find which bands differ most between them.

3. **More vegetable crops.** Kailan next, then Arabidopsis, with the same pipeline.

4. **Raman and image fusion.** Classify the fused feature vectors, and compare image only, Raman only and fused.

5. **Paper.** Aim to submit in January or February.

APPENDIX

What each term means

Term	Meaning
Patch	A square cut from the image; now 224 × 224 pixels, being tuned from 8 to 224
Stride	How many pixels the patch window moves each step; now 4, being tuned (2 or 4)
Epoch / run	Epoch: one pass of training over all training patches. Run: one full training from a fresh random start
Plant vote	All patches of a plant vote; the most common label is the plant’s answer
White-light normalisation	Each image divided by that day’s healthy-plant average, so stress shows as a difference
Accuracy	Share of plants given the correct condition. 0.25 is random guessing
Precision	When the model names a condition, how often it is right: TP / (TP + FP)
Recall	Of the plants truly in a condition, how many it finds: TP / (TP + FN)
F1	One score balancing precision and recall, 2PR / (P + R); 1 is perfect
TP / FP / FN / TN	For one condition: correctly flagged / wrongly flagged / missed / correctly left out
Confusion matrix	Real condition in rows, model’s answer in columns; the diagonal is correct
Stressed vs control	Same answers grouped into two: healthy, or any of the three stresses